Chapter 9

1. What index value does the third element of an array have?

The third element of an array has an index value of **2** because arrays start counting at 0.

First element → index 0

Second element → index 1

Third element → index 2

2. Write the declaration for an for an array named `quantities` that stores 20 integers.

```
int[] quantities = new int[20];
```

3. Write a declaration for an array named heights storing the numbers 1.65, 2.15, and 4.95.

```
double[] heights = {1.65, 2.15, 4.95};
```

4. Write a for-each statement that displays the integer values stored in an array named grades.

```
for (int grade : grades) {
    System.out.println(grade);
     }
```

6. How does passing an entire array to a method differ from passing a single element of the array?

Passing an entire array to a method allows the method to access and modify all the elements in the array. The method receives a reference to the array, not a copy of all its values.

8. What output is displayed by the statements below?

```
String name - "Elaine";
System. out. print In(name. charAt (3));
```

10. Give an example of when a dynamic array might be a better structure choice over an array.

A dynamic array is a better choice when the amount of data can change during the program. For example, a shopping cart in an online store may continue growing as users add more items. A regular array has a fixed size, but a dynamic array can automatically resize when needed.

11. How does the ArrayList indexOf() method determine equality between the object passed to the method and an element in the array?

The ArrayList indexOf() method determines the equality by using the object's .equals() method. It compares the object passed into indexOf() with each element in the ArrayList until it finds a match. If the objects are equal, it returns the index of that element.

12. How can the values of wrapper class objects be compared?

The values of wrapper class objects can be compared using the .equals() method. This method checks if the actual values inside the objects are the same, instead of comparing their memory locations.